

Problem

In a certain electronic device, a three-stage amplifier is desired, whose overall voltage gain is 42 dB. The individual voltage gains of the first two stages are to be equal, while the gain of the third is to be one-fourth of each of the first two. Calculate the voltage gain of each.

Step-by-step solution

Step 1 of 3

Let the individual gain of the first two stages is K .

$$A_1 = K$$

$$A_2 = K$$

The individual gain of the third stage is one-fourth of the gain of first and second stage gains.

$$A_3 = \frac{K}{4}$$

Calculate the overall gain of the circuit.

$$\begin{aligned} A &= A_1 A_2 A_3 \\ &= (K)(K)\left(\frac{K}{4}\right) \\ &= \frac{K^3}{4} \end{aligned}$$

Step 2 of 3

Convert the overall gain in decibels.

$$\begin{aligned} A(\text{dB}) &= 20 \log A \\ &= 20 \log \frac{K^3}{4} \end{aligned}$$

Substitute 42 dB for $A(\text{dB})$.

$$20 \log \frac{K^3}{4} = 42$$

$$\log \frac{K^3}{4} = \frac{42}{20}$$

$$\log \frac{K^3}{4} = 2.1$$

$$K^3 = 4 \times 10^{2.1}$$

$$K^3 = 503.57$$

$$K = (503.57)^{\frac{1}{3}}$$

$$K = 7.959$$

Step 3 of 3

The individual gain of the first two stages is,

$$A_1 = 7.959 \text{ V/V}$$

$$A_2 = 7.959 \text{ V/V}$$

The individual gain of the third stage is one-fourth of the gain of first and second stage gains.

$$A_3 = \frac{K}{4}$$

$$= \frac{7.959}{4}$$

$$= 1.989 \text{ V/V}$$

Thus, the individual gains of the three stage amplifier are,

$$A_1 = 7.959 \text{ V/V}, A_2 = 7.959 \text{ V/V}, A_3 = 1.989 \text{ V/V}.$$